# Endpoint DLP for macOS — Microsoft Purview

Implementation Solution Guide

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# Executive Summary — Endpoint DLP for macOS (Microsoft Purview)

Data Loss Prevention programs are often built for Windows and the cloud — and leave **Macs as the blind spot**. The same sensitive content that is blocked from leaving a Windows PC can walk off a Mac over a USB drive, a browser upload, a network copy, the clipboard, or a print job. That is a compliance gap and a real data-risk exposure. The fix does not require a new product: **Microsoft Purview Endpoint DLP** already extends to **macOS**, in the tenant and policy engine you run today.

This solution delivers macOS Endpoint DLP end to end: **onboarding** Macs into Purview device management (via **Intune** or **Jamf Pro**), configuring the shared **Endpoint DLP settings**, and building a **DLP policy** that monitors and controls the key **egress channels** on the Mac — piloted in **simulation / test (Audit)** first, validated in **Activity Explorer**, and only then **enforced** on a controlled scope. It is written for an administrator with no prior macOS Endpoint-DLP experience.

## What you get

• A plain-language **Solution Overview** of how macOS Endpoint DLP works — onboarding, the system/network extensions and Full Disk Access, the egress channels, and the simulation-first model.

• A **Requirements Guide** with verify / implement-if-missing for every prerequisite (the category Purview requirements — licensing, roles, toolchain, classification intent, stakeholders, pilot — are referenced, not repeated).

• An **Implementation Guide** of precise, ordered **Purview + Intune + Security & Compliance PowerShell **steps: onboard the Macs, set Endpoint DLP settings, create the policy** in test/simulation**, pilot, validate, then enforce — each object named, with pass/fail gates and a one-command rollback.

• Five **PowerShell scripts** for readiness discovery, safe policy creation (`-WhatIf`, test-mode only), validation, Activity-Explorer reporting, and an onboarding/scope health check.

• **Testing** and **Troubleshooting** guides for a confident rollout.

## What this delivers

macOS endpoints **onboarded** to Purview and **watched** across cloud/browser upload, USB, network share, clipboard, print, and restricted-app access — **audited in simulation**, **validated** with evidence, and **enforced** (Block) on the channels compliance approves, in waves. The Mac blind spot is closed inside the DLP program you already own.

## The deployment model

**Assess** → **Design** → **Configure** → **Pilot (simulate)** → **Validate** → **Expand (enforce)** — build onboarding and the policy, prove them silently on a handful of Macs, then enforce and roll out in waves.

## An honest note

macOS Endpoint DLP support is a **subset of Windows** and is **evolving** (limited to the three latest major macOS versions); onboarding requires real endpoint work (extension approval + Full Disk Access via MDM). Enforcement is **high-impact** — hence simulation-first with a documented rollback. **Which** data is regulated and **what** must be blocked are **compliance, records, and legal** determinations; this package provides the technical configuration and the evidence, not the determination — and flags **"verify your licensing / tenant / macOS version"** wherever availability is plan- or version- dependent.

### Outcome

A macOS fleet inside your **Microsoft Purview** DLP program — onboarded, monitored across the key egress channels, piloted in simulation, validated, and enforced on a controlled scope: a closed blind spot that is configured, evidenced, and defensible.

## 1. Solution Overview — Endpoint DLP for macOS (Microsoft Purview)

### What this solution is

This solution extends **Microsoft Purview Endpoint Data Loss Prevention (Endpoint DLP)** to **macOS devices**. It covers the whole path: **onboarding** Macs into Microsoft Purview device management, configuring the **Endpoint DLP settings** and a **DLP policy** that monitors and controls **egress channels** on the Mac (cloud/browser upload, USB removable media, network share, clipboard, print, and access by unallowed apps) — piloting the policy in **simulation / test mode** first, validating what it would catch, and only then moving to **enforce** on a controlled scope.

Mac endpoints are frequently the **blind spot** in a DLP program: the same sensitive content that is blocked from leaving a Windows PC can walk out on a Mac over USB, a browser upload, or a network copy. This package closes that gap in Purview, the same tenant and policy engine you already use — no third-party endpoint DLP product.

It is **vendor-neutral** on the management plane: macOS onboarding is authored for both **Microsoft Intune** and **Jamf Pro** at the pattern level, with the Intune path given step-by-step and the Jamf equivalent named and deferred to Jamf's documentation for its console specifics.

### The macOS Endpoint DLP model in one paragraph

Endpoint DLP works by **onboarding** a device so a management/monitoring agent reports device activity to Microsoft Purview. On macOS this agent relies on macOS **system and network extensions**, a **Full Disk Access** grant, and an **onboarding package** delivered by your MDM (Intune or Jamf). Once a Mac is onboarded and appears in Purview **device onboarding**, a **DLP policy** scoped to the **Devices (Endpoint)** location applies to it. The policy's **rules** detect sensitive content (by sensitive information type, sensitivity label, or other conditions) and apply **endpoint restrictions** to egress activities — **Audit**, **Warn/Block with override**, or **Block**. Global behavior (which browsers are approved, which apps are "unallowed", which cloud domains are sensitive/approved, path exclusions) lives in **Endpoint DLP settings** and is shared by all endpoint policies.

> ***Licensing / plan note (verify your licensing/tenant).*** *Endpoint DLP is part of* ***Microsoft Purview Data Loss Prevention*** *and is licensed with* ***Microsoft 365 E5 / E5 Compliance*** *(or equivalent). macOS Endpoint DLP support and the exact set of monitored activities are* ***evolving and version-dependent*** *— Microsoft supports Endpoint DLP on the* ***three latest major macOS releases****. Confirm current macOS support, licensing, and per-activity availability against Microsoft's public documentation and your tenant before relying on any specific control.*

## Egress channels this solution targets on macOS

macOS Endpoint DLP can monitor and restrict a **subset** of the endpoint activities available on Windows. The channels this solution configures (each **verify current support for your macOS version**):

- **Cloud / browser upload** — uploading sensitive files to a **cloud service domain** from a browser,

and uploading from an **unallowed / non-approved browser**. Coverage of **service domains** and **sensitive service domains** on non-Edge browsers (Chrome, Firefox) requires the **Microsoft Purview browser extension**; Edge is supported without an add-on.

- **USB / removable media** — copying sensitive files to a USB or other removable drive.
- **Network share** — copying sensitive files to a network share.
- **Clipboard** — copying sensitive content to the clipboard (paste into another app/browser).
- **Print** — printing sensitive files.
- **Unallowed apps / restricted app groups** — access to sensitive files by apps you designate as

restricted (e.g. personal cloud-sync or messaging apps).

For each channel the DLP rule sets an **action**: **Audit** (log only — the safe pilot default), **Block with override / Warn** (user can justify), or **Block** (hard stop). This solution pilots at **Audit / simulation** and only tightens to Block after validation.

## What is authored exactly vs deferred

- **Exact navigation is given** for the **Microsoft Purview portal** (`purview.microsoft.com`) —

**Device onboarding**, **Endpoint DLP settings**, and **Data Loss Prevention** → **Policies** — and for the **Intune** onboarding path and **Security & Compliance PowerShell** (`Connect-IPPSSession`) automation.

- **Jamf Pro** onboarding is integrated at the **pattern level** — where a step (deploy onboarding

package, approve system/network extensions, grant Full Disk Access via a PPPC/TCC profile) is a Jamf console action, this solution names the action and defers the Jamf specifics to Jamf's documentation.

- **Records/retention/legal/regulatory determinations** — *which* data is regulated, *what* must be

blocked vs merely audited, and *whether* a given activity is permissible — are **deferred to the organization's compliance, privacy, records, and legal functions**. This solution delivers the **technical configuration and the evidence**, not the policy determination.

- **Purview is evolving and licensing-dependent** — where a control's availability on macOS is

version- or plan-dependent, this package flags **"verify your licensing / tenant / macOS version"** rather than asserting it.

## Benefits

• **Closes the Mac blind spot** — the same Purview DLP engine now watches egress on macOS, not just

Windows and cloud.

• **One policy plane** — Macs are governed by DLP policies in the tenant you already run; no separate

endpoint-DLP product.

• **Safe by construction** — every policy is piloted in **simulation / test (Audit)** first, validated

in Activity Explorer, then enforced — high-impact blocking is never flipped on blind.

• **Evidenced** — Activity Explorer and the provided reporting script show exactly what the policy

would catch (and later, blocks), so the control is **defensible**.

• **Automatable & safe** — Security & Compliance PowerShell / Graph scripts are read-only or run in

**-WhatIf / test mode** by default, with no credentials in code.

## Limitations (be honest with buyers)

• **macOS support is a subset and evolving.** Not every Windows Endpoint DLP activity exists on macOS,

and support is limited to the **three latest major macOS versions** — **verify current coverage**.

• **Onboarding is real endpoint work.** macOS requires **system/network extension approval**, **Full

Disk Access**, and an** onboarding package** via MDM — a misconfigured extension/TCC profile means the Mac never reports and the policy silently does nothing.

• **Determinations are not ours.** What to block, what is regulated, and whether an activity is allowed

are **compliance/records/legal** decisions — this package configures and evidences, it does not decide.

• **Enforcement is high-impact.** A Block rule can interrupt legitimate work — hence **simulation →

Audit → Block** with pass/fail gates and a documented rollback.

## The standard deployment model

> ***Assess*** *the Mac fleet and in-scope data →* ***Design*** *the onboarding + policy + endpoint settings →* ***Configure*** *onboarding and a DLP policy in* ***test/simulation (Audit)*** *→* ***Pilot*** *on a handful of onboarded Macs →* ***Validate*** *in Activity Explorer against pass/fail gates →* ***Expand*** *and only then move rules to* ***Block (enforce)*** *in waves. Pattern:* ***Assess*** *→* ***Design*** *→* ***Configure*** *→* ***Pilot (simulate/test)*** *→* ***Validate*** *→* ***Expand (enforce)****, with explicit pass/fail gates and a documented rollback.*

# 2. Requirements Guide — Endpoint DLP for macOS (Microsoft Purview)

Verify **every** requirement before configuring. Each entry: **Description · Why required · How to verify · How to implement if missing.** Category-wide Purview prerequisites (a tenant with the right **compliance licensing**, least-privilege **Purview roles**, the **Security & Compliance PowerShell / Graph toolchain**, **in-scope data/classification**, **compliance/records/privacy/legal stakeholders**, and a **pilot scope + change control with simulation/test mode**) are in `global-requirements-purview.md` — confirm those **first**; this guide adds the macOS Endpoint DLP specifics and does not duplicate the six.

> ***Object names used throughout (use verbatim):*** *DLP policy* `Endpoint DLP - macOS - Pilot` *(and* `Endpoint DLP - macOS - Prod`*); DLP rule* `macOS Egress - Sensitive Content`*; macOS onboarding configuration (Intune)* `Purview DLP - macOS Onboarding`*; restricted app group (Endpoint DLP settings)* `macOS - Restricted Apps`*; approved/sensitive cloud-domain intent recorded in* ***Endpoint DLP settings → Sensitive service domains / Browser and domain restrictions****; policy* ***Mode*** *progression* `TestWithoutNotifications` → `TestWithNotifications` → `Enable`*.*

## 2.1 Endpoint DLP licensing (E5 / E5 Compliance) — category G1

**Description:** The tenant is licensed for **Microsoft Purview Data Loss Prevention including Endpoint DLP** — **Microsoft 365 E5** or the **E5 Compliance** add-on (or equivalent). **Why required:** Endpoint DLP is a premium DLP capability; without the plan the **Devices** location and endpoint restrictions are unavailable. *(Verify your licensing/tenant.)* **How to verify:** Purview portal → **Data Loss Prevention** is available and a policy can select the **Devices** location; M365 admin center shows the subscription/add-on. **How to implement if missing:** Assign/purchase the required licensing before starting.

## 2.2 Least-privilege Purview / DLP roles (category G2)

**Description:** An operator in a least-privilege Purview role that can manage DLP — e.g. **Compliance Administrator** or **Compliance Data Administrator** / **DLP Compliance Management**, plus **Information Protection** where labels are involved. Device onboarding + Activity Explorer viewing require the corresponding compliance roles. **Why required:** Purview uses fine-grained role groups; creating policies, editing Endpoint DLP settings, and onboarding devices each require the matching role. **How to verify:** Purview portal → **Settings** → **Roles & scopes / Permissions** shows the role membership; the operator can open **Data Loss Prevention** and **Device onboarding**. **How to implement if missing:** Assign the least-privilege role group under change control (Global Administrator only where genuinely required).

## 2.3 Security & Compliance PowerShell + Graph toolchain (category G3)

**Description:** A workstation with **Security & Compliance PowerShell** (`Connect-IPPSSession` via the **ExchangeOnlineManagement** module) and, for the onboarding/health helper, **Microsoft Graph PowerShell** — signed in as a modern-auth / MFA-capable admin. **Why required:** The DLP policy/rule automation and reporting run through Security & Compliance PowerShell; the onboarding scope check uses Graph against Intune. **How to verify:** `Connect-IPPSSession` and `Connect-MgGraph` both succeed; `DLPMac-Discovery.ps1` runs read-only. **How to implement if missing:** `Install-Module ExchangeOnlineManagement` and `Install-Module Microsoft.Graph`; confirm modern-auth sign-in before running steps.

## 2.4 Supported macOS versions in scope

**Description:** The target Macs run a **supported macOS release** — Microsoft supports Endpoint DLP on the **three latest major macOS versions** *(verify current support)*. **Why required:** The DLP system/network extensions and monitored activities are tied to supported macOS versions; older releases are unsupported and may not report. **How to verify:** Confirm the fleet's macOS versions against Microsoft's current supported list; `DLPMac-Onboarding-Check.ps1` reports macOS device OS versions from Intune. **How to implement if missing:** Update Macs to a supported macOS version before onboarding.

## 2.5 An MDM to deliver onboarding (Intune or Jamf Pro)

**Description:** A managed path to deploy the **onboarding package**, the **system/network extension** approvals, and the **Full Disk Access (PPPC/TCC)** profile to Macs — **Microsoft Intune** (authored step-by-step) or **Jamf Pro** (pattern-level, deferred to Jamf docs). **Why required:** macOS will not load the DLP extensions or grant Full Disk Access silently — these are delivered as MDM configuration profiles; without MDM you cannot reliably onboard at scale. **How to verify:** Macs are enrolled in Intune (or Jamf) and can receive configuration profiles; `DLPMac-Onboarding-Check.ps1` lists managed macOS devices. **How to implement if missing:** Enroll the Macs in Intune or Jamf first (a separate prerequisite), then return here.

## 2.6 macOS devices onboarded to Microsoft Purview device management

**Description:** The pilot Macs are **onboarded** and appear in Purview **Device onboarding** as reporting devices (this is what makes an endpoint-scoped DLP policy apply to them). **Why required:** A DLP policy scoped to **Devices** only affects **onboarded** devices; an un-onboarded Mac is invisible to Endpoint DLP and the policy silently does nothing on it. **How to verify:** Purview portal → **Settings** → **Device onboarding** → **Devices** lists the Macs as onboarded/active *(verify the exact portal path for your tenant)*. There is no supported Security & Compliance PowerShell cmdlet that enumerates onboarded devices — confirm in the portal. **How to implement if missing:** Onboard via Intune (Implementation Phase 1) or Jamf; confirm the Mac reports in device onboarding before scoping the policy.

## 2.7 macOS system/network extension + Full Disk Access approved

**Description:** The macOS **system extension** and **network (content-filter) extension** are **approved**, and **Full Disk Access** is granted to the DLP/Defender agent via a **PPPC/TCC** configuration profile. **Why required:** Without extension approval and Full Disk Access, the Mac cannot inspect file activity or egress — the agent installs but reports nothing, so the policy appears to "not work". **How to verify:** On a pilot Mac, **System Settings** → **Privacy & Security** shows the extension allowed and Full Disk Access granted; the device reports activity in Activity Explorer after a test action. **How to implement if missing:** Deploy the system-extension allow-list, network-extension, and PPPC/Full-Disk-Access profiles via Intune (Phase 1) or Jamf; these must be **pushed by MDM**, not clicked per device, for a silent rollout.

## 2.8 The Microsoft Purview browser extension for non-Edge browsers (cloud egress)

**Description:** For **cloud/service-domain** coverage in **Chrome / Firefox** on macOS, the **Microsoft Purview extension** is deployed; **Microsoft Edge** is supported without an add-on. **Why required:** Sensitive-service-domain and "unallowed browser" controls on non-Edge browsers depend on the

extension; without it, uploads via Chrome/Firefox are not evaluated for those controls. **How to verify:** The extension is present/forced in the target browsers (via Intune/Jamf browser policy); a test upload from Chrome/Firefox is evaluated. **How to implement if missing:** Force-install the Purview extension in the in-scope browsers, or restrict cloud-egress testing to Edge for the initial pilot.

## 2.9 In-scope sensitive data + classification intent (category G4)

**Description:** The **sensitive content** the policy detects — **sensitive information types (SITs)**, **sensitivity labels**, and/or keyword/EDM conditions — agreed with data owners, and which **egress channels** matter for the Mac fleet. **Why required:** A DLP rule keys off *what* is sensitive; without an agreed detection basis the policy either over-triggers (noise) or misses (gap). The classification determination is a governance input. **How to verify:** A named list of SITs/labels/conditions and the target egress channels exists and is signed off by data owners. **How to implement if missing:** Work with compliance/data owners to define the detection basis (category G4/G5) before authoring the rule.

## 2.10 Endpoint DLP settings baseline (restricted apps, domains, exclusions)

**Description:** The tenant-wide **Endpoint DLP settings** relevant to macOS — **restricted app groups** (`macOS - Restricted Apps`), **sensitive/approved service domains**, **browser & domain restrictions**, and **file-path exclusions** — are reviewed and set to the pilot intent. **Why required:** These global settings are shared by all endpoint DLP policies; they control which browsers/apps/domains the rules act on. Wrong exclusions = false negatives; missing restricted-app groups = the app rule can't reference them. **How to verify:** Purview portal → **Data Loss Prevention** → **Endpoint DLP settings** reflects the intended app groups, domains, and exclusions; `DLPMac-Validation.ps1` flags a rule that references a group not present. **How to implement if missing:** Configure the settings in Implementation Phase 2 before enabling the rule's app/domain restrictions.

## 2.11 Pilot scope + change control with simulation/test mode (category G6)

**Description:** A small **pilot set** of onboarded Macs, a **change-control** window, and use of the DLP policy's **test/simulation mode** (`TestWithoutNotifications` → `TestWithNotifications`) **before** `Enable` (enforce). Endpoint DLP blocking is high-impact. **Why required:** Every Purview solution follows **Assess** → **Design** → **Configure** → **Pilot (simulate)** → **Validate** → **Expand (enforce)**; a Block rule can interrupt legitimate work, so it is validated in simulation/Audit first. **How to verify:** A named pilot Mac set, an approval window, and the policy created in a **test** mode (not `Enable`) exist; `DLPMac-Discovery.ps1` reports each endpoint policy's Mode. **How to implement if missing:** Define the pilot and agree change control; create the policy in `TestWithoutNotifications` first (Implementation Phase 3).

## Checklist

- Endpoint DLP licensing (E5 / E5 Compliance) confirmed (2.1 / category G1)
- Least-privilege Purview/DLP role on the operator (2.2 / category G2)
- Security & Compliance PowerShell + Graph toolchain working (2.3 / category G3)
- Fleet on a supported macOS version (three latest major) (2.4)
- MDM (Intune or Jamf) able to deliver onboarding + profiles (2.5)

- Pilot Macs onboarded and visible in Purview device onboarding (2.6)
- System/network extension approved + Full Disk Access granted (2.7)
- Purview browser extension deployed for Chrome/Firefox cloud egress (2.8)
- In-scope SITs/labels/conditions + target egress channels agreed (2.9 / category G4)
- Endpoint DLP settings baseline (restricted apps, domains, exclusions) set (2.10)
- Pilot scope + change control + policy in test/simulation mode (2.11 / category G6)

# Global Requirements — Microsoft Purview Solutions

These requirements apply to **every** solution in the **Purview** category (implementing and managing Microsoft Purview, sensitivity & retention labels, DLP, data lifecycle & records, insider risk, communication compliance, audit, data security posture management, and data governance). Verify each before starting a solution's implementation steps.

> ***Vendor boundary (applies to every Purview solution).*** *The **Microsoft Purview portal** (purview.microsoft.com), the **Microsoft 365 admin center**, **Security & Compliance PowerShell** (`Connect-IPPSSession`), and **Microsoft Graph** are authored at exact, current click-by-click / cmdlet detail from Microsoft's public documentation. Microsoft Purview features are heavily **licensing-/plan-dependent** (many require **Microsoft 365 E5** or the **E5 Compliance** add-on; some are in preview) and the portal is **actively evolving** — capability availability and portal paths are **cited or flagged "verify your licensing/tenant"**, never asserted or invented. **Regulatory, records-retention, and legal determinations** (retention schedules, records file plans, legal holds, insider-risk case actions, regulatory applicability) are **deferred to the organization's compliance, records, privacy, and legal functions** — the solutions provide the technical configuration method and evidence, not the determination. Automation is **PowerShell** (Security & Compliance PowerShell / Graph) — never with hardcoded credentials or secrets.*

## 1. Microsoft 365 tenant with the required Purview/compliance licensing

- **Description:** A Microsoft 365 tenant with the **compliance licensing** the capability requires

(E3 for baseline; **E5 / E5 Compliance** for advanced DLP, auto-labeling, insider risk, communication compliance, Audit Premium, DSPM, etc.).

- **Why required:** Most Purview capabilities are plan-gated; the feature won't be available without the

right plan.

- **How to verify:** In the M365 admin center confirm the subscription/add-ons; check the feature is

visible in the Purview portal.

- **How to implement if missing:** Assign/purchase the required licensing before starting.

## 2. Purview roles (least-privilege) and portal access

- **Description:** Access with the least-privilege **Purview / compliance role** the solution names

(e.g. Compliance Administrator, Compliance Data Administrator, Information Protection, Records Management, Insider Risk, eDiscovery, Audit roles), with Global Administrator only where required.

- **Why required:** Purview uses fine-grained role groups; configuration requires the matching role.
- **How to verify:** Purview portal → **Roles & scopes / Permissions** — confirm the role membership.
- **How to implement if missing:** Assign the least-privilege role group under change control.

## 3. Security & Compliance PowerShell + Graph toolchain

- **Description:** A workstation with **Security & Compliance PowerShell** (`Connect-IPPSSession` via the

ExchangeOnlineManagement module) and **Microsoft Graph PowerShell**, modern-auth / MFA-capable admin.

- **Why required:** Configuration, automation, and the provided scripts run through these.
- **How to verify:** `Connect-IPPSSession` and `Connect-MgGraph` succeed with the admin account.
- **How to implement if missing:** Install the modules and confirm modern-auth sign-in before running steps.

## 4. Data sources & scope in scope

- **Description:** The **workloads and data sources** in scope (Exchange, SharePoint, OneDrive, Teams,

endpoints/devices, and — for data governance — the broader data estate) and a **data-classification / sensitivity** intent to drive labels, DLP, and retention.

- **Why required:** Purview policies key off the in-scope workloads and the classification scheme.
- **How to verify:** Confirm the in-scope workloads and the classification/sensitivity approach with

stakeholders.

- **How to implement if missing:** Define the scope and classification intent before authoring policies.

## 5. Compliance, records, privacy & legal stakeholders

- **Description:** Engagement with the **compliance, records-management, privacy, security, and legal**

stakeholders whose policies the configuration must satisfy (retention schedules, records file plans, regulatory obligations, insider-risk and communication-compliance policy, legal hold).

- **Why required:** Purview configuration must align to enterprise policy and legal/records requirements,

not technical defaults; several capabilities have HR/legal sensitivity.

- **How to verify:** Confirm the applicable policies, the records schedule, and named approvers.
- **How to implement if missing:** Identify the owners and capture the requirements before configuring.

## 6. A pilot scope and change control

• **Description:** A **pilot scope** (users/sites/policies in simulation or a limited ring) to validate

on before tenant-wide enforcement, and a **change-control** process — most Purview policies support a **simulation / test / disabled** mode that MUST be used before enforcement.

• **Why required:** Every solution follows an **Assess → Design → Configure → Pilot (simulate) → Validate

→ Expand (enforce)** model; DLP, labels, retention, and insider-risk changes are high-impact.

• **How to verify:** Confirm a named pilot scope, that simulation/test mode is available, and a change

window/approval path.

• **How to implement if missing:** Define the pilot and agree change control before enforcing anything.

*The deployment model across every Purview solution is the same: **Assess** → **Design** → **Configure** → **Pilot (simulate/test)** → **Validate** → **Expand (enforce)**, with explicit pass/fail gates and a documented rollback. Automation is delivered as **PowerShell** (Security & Compliance PowerShell / Graph) — never with hardcoded credentials or secrets. Regulatory/records/legal determinations are deferred to the relevant governance functions.*

# 3. Implementation Guide — Endpoint DLP for macOS (Microsoft Purview)

Follow these steps **in order**. The **Microsoft Purview portal**, the **Intune** onboarding path, and the **Security & Compliance PowerShell** automation are given click-by-click / cmdlet-exact. **Jamf Pro** onboarding is named and deferred to Jamf's documentation for its console specifics. Category Purview prerequisites (licensing, roles, toolchain, classification intent, stakeholders, pilot/change control) are assumed confirmed (Requirements + `global-requirements-purview.md`).

Consoles used:

• **Microsoft Purview portal** — `https://purview.microsoft.com` (Device onboarding, Endpoint DLP

settings, Data Loss Prevention → Policies)

• **Microsoft Intune admin center** — `https://intune.microsoft.com` (macOS onboarding + extension /

Full Disk Access profiles)

• **Jamf Pro** — deferred to Jamf documentation for the equivalent profiles

• **Security & Compliance PowerShell** — `Connect-IPPSSession`

> ***Model:*** *Onboard the Macs, configure the endpoint settings, create the DLP policy **in test/simulation (Audit)**, pilot on a few onboarded Macs, validate in Activity Explorer, then move rules to **Block (enforce)** in waves. **Assess** → **Design** → **Configure** → **Pilot (simulate)** → **Validate** → **Expand (enforce).***
>
> ***Artifact names (use verbatim):*** *onboarding config (Intune)* `Purview DLP - macOS Onboarding`*; DLP policy* `Endpoint DLP - macOS - Pilot` *(later* `Endpoint DLP - macOS - Prod`*); DLP rule* `macOS Egress - Sensitive Content`*; restricted app group* `macOS - Restricted Apps`*.*

> ***Risk + rollback (read first):*** *Endpoint DLP* ***Block*** *actions can interrupt legitimate work (blocked USB copy, blocked upload, blocked print). This solution therefore creates the policy in* `TestWithoutNotifications` *(silent simulation / Audit) and only advances to* `Enable` *after validation.* ***Rollback:*** *set the policy* ***Mode*** *back to* `TestWithoutNotifications` *(or* `Disable`*) to stop enforcing immediately (*`Set-DlpCompliancePolicy -Mode`*); un-scope by removing the* ***Devices*** *location; fully stop by disabling the policy. Onboarding is reversible by* ***off-boarding*** *the device in Purview / removing the MDM profiles. Regulatory/records/legal determinations remain with the relevant governance functions — this guide configures the technical control.*

## Phase 1 — Onboard macOS devices to Microsoft Purview

### *Step 1 — Get the macOS onboarding package from Purview*

1. Purview portal → **Settings** → **Device onboarding** *(verify the exact path for your tenant; the onboarding blade has moved between the compliance portal and the new Purview portal)*.

1. Choose **macOS** and the **deployment method** — **Intune** or **Jamf** — and **download the onboarding package** (a configuration payload / `.mobileconfig` / onboarding blob).

1. Note that onboarding for Endpoint DLP on macOS uses the same device-onboarding pipeline as Microsoft

Defender for Endpoint; if MDE is already deployed on the Mac, the extensions may be shared — **verify against current Microsoft guidance for your tenant**.

### *Step 2 — Deploy onboarding + required profiles via Intune*

Create these in Intune → **Devices** → **macOS** → **Configuration** (name the onboarding one `Purview DLP - macOS Onboarding`) and assign to the **pilot Mac group**:

1. **Onboarding package** — upload the payload from Step 1 as a **configuration profile** (custom / `.mobileconfig`).

1. **System extension** — a **System Extensions** profile that **allows** the DLP/Defender extension (team identifier / bundle IDs per Microsoft's macOS deployment docs).

1. **Network extension** — a **content-filter / network extension** profile so egress can be inspected.

2. **Full Disk Access (PPPC/TCC)** — a **Privacy Preferences Policy Control** profile granting **Full Disk Access** to the agent so it can see file activity.

1. **(Optional) Background/Notifications** profiles per Microsoft's macOS guidance.

2. Assign all profiles to the **pilot Mac group** and let them apply.

> ***Jamf path (deferred):*** *deploy the same four payloads — onboarding package,* ***System Extensions*** *allow,* ***Content Filter / network*** *extension, and* ***PPPC (Full Disk Access)*** *— via Jamf* ***Configuration Profiles****. The payload contents are Microsoft's; the Jamf console steps are deferred to Jamf's docs.*

### *Step 3 — Confirm the Macs are onboarded*

1. Purview portal → **Settings** → **Device onboarding** → **Devices** — the pilot Macs appear as

**onboarded/active** *(verify path)*.

1. On a pilot Mac, **System Settings** → **Privacy & Security** shows the extension **allowed** and **Full Disk Access** granted.
1. Run `DLPMac-Onboarding-Check.ps1` — it reports managed macOS devices, OS versions, and whether the onboarding/extension configuration profiles are present in Intune.
1. **Pass/fail gate:** at least the pilot Macs are onboarded and show the extension + Full Disk Access.

Do **not** proceed until Macs report — a policy on an un-onboarded Mac does nothing.

## Phase 2 — Configure Endpoint DLP settings (shared, tenant-wide)

### *Step 4 — Set the Endpoint DLP settings baseline*

Purview portal → **Data Loss Prevention** → **Endpoint DLP settings**. Review/set (to the pilot intent agreed with data owners — determinations remain theirs):

1. **Restricted apps and app groups** — create the app group `macOS - Restricted Apps` listing the macOS apps you want to restrict access to sensitive files (e.g. personal cloud-sync / messaging apps, by bundle ID/name). The DLP rule references this group.
1. **Browser and domain restrictions to sensitive data** — set **unallowed browsers** (so sensitive files can't upload via non-approved browsers) and confirm **Microsoft Edge** is the approved browser; deploy the **Purview browser extension** for Chrome/Firefox (Requirement 2.8).
1. **Sensitive service domains** — define the **service domain groups** (approved vs sensitive) that the cloud-egress control acts on.
1. **File path exclusions (macOS)** — add any macOS paths to exclude from monitoring (to cut noise), being careful not to exclude paths that would create a gap.

> *These settings are **shared by every endpoint DLP policy** — change them under change control.* `DLPMac-Validation.ps1` *flags a rule that references the* `macOS - Restricted Apps` *group if the group is absent.*

## Phase 3 — Create the DLP policy in simulation/test (Audit) mode

### *Step 5 — Create the endpoint-scoped DLP policy (test mode)*

1. **Portal path:** Purview → **Data Loss Prevention** → **Policies** → **Create policy** → choose the data categories/custom, name it `Endpoint DLP - macOS - Pilot`, and select the **Devices** location **only** (turn off Exchange/SharePoint/OneDrive/Teams for this Mac-focused pilot).
1. On **Policy mode**, choose **Run the policy in simulation mode** (optionally with policy tips off) — this is the `TestWithoutNotifications` mode: it **audits/simulates** without user-visible tips or blocks.

1. **Automation (preferred, safe):** run `DLPMac-Set-Policy.ps1` — default **-WhatIf** (previews the `New-DlpCompliancePolicy` / `New-DlpComplianceRule` it would run); add **-Commit** to create the policy **in** `TestWithoutNotifications` and the rule with **Audit** endpoint restrictions:

```
New-DlpCompliancePolicy -Name 'Endpoint DLP - macOS - Pilot' -EndpointDlpLocation All -Mode TestWithout
```

The script never creates the policy in `Enable` mode — enforcement is a deliberate later step.

### *Step 6 — Add the egress rule (Audit actions first)*

1. In the policy, **Create rule** `macOS Egress - Sensitive Content`.
2. **Conditions:** content is sensitive — the agreed **SIT(s) / sensitivity label(s) / conditions** (Requirement 2.9).
1. **Endpoint restrictions (start at Audit):** set each in-scope channel to **Audit** for the pilot:

- **Upload to a cloud service / access by unallowed browser** → **Audit**
- **Copy to a USB removable device** → **Audit**
- **Copy to a network share** → **Audit**
- **Copy to clipboard** → **Audit**
- **Print** → **Audit**
- **Access by restricted apps** (`macOS - Restricted Apps`) → **Audit**

*(Only include channels supported on your macOS version — verify current support.)*

1. Save the rule. `DLPMac-Set-Policy.ps1 -Commit` creates this rule with **Audit** restrictions via `New-DlpComplianceRule -EndpointDlpRestrictions` — never Block on first creation.

### *Step 7 — Confirm the policy is scoped and in test mode*

1. Run `DLPMac-Discovery.ps1` — it lists the endpoint-scoped policy, confirms **Mode = TestWithoutNotifications** (flags anything in `Enable`), and lists the rule's egress restrictions.
1. Run `DLPMac-Validation.ps1` — confirms the policy targets **Devices**, is in a **test** mode, and the rule includes the intended channels + references a present restricted-app group.
1. **Pass/fail gate:** policy exists, **Devices**-scoped, **Mode = TestWithoutNotifications**, rule present with **Audit** actions. Do not advance until this passes.

## Phase 4 — Pilot, validate, then enforce

### *Step 8 — Drive test activities on a pilot Mac and validate*

1. On an onboarded pilot Mac, perform benign test egress with **non-production test content that matches the SIT** (e.g. a file with dummy test data): copy to USB, copy to a network share, upload from a browser, copy to clipboard, print, open with a restricted app.
1. Purview portal → **Activity Explorer** (or run `DLPMac-Report.ps1`, which uses

`Export-ActivityExplorerData`) — the simulated/audited **Endpoint DLP** events for the Mac appear with the activity, app, and matched rule.

1. **Pass/fail gate (simulation):** every driven channel produces an **audited** event attributed to

`Endpoint DLP - macOS - Pilot` / `macOS Egress - Sensitive Content`, and no unexpected volume of false positives. If a channel produces nothing, revisit onboarding/extensions (Phase 1) or the channel's macOS support before enforcing.

### *Step 9 — Advance to notifications, then enforce (Block) — deliberately, in scope*

Only after Step 8 passes and **compliance/records/legal sign off** the enforcement intent:

1. `TestWithNotifications` — move the policy to test **with** policy tips so pilot users see

warnings (still no block): `Set-DlpCompliancePolicy -Identity 'Endpoint DLP - macOS - Pilot' -Mode TestWithNotifications`. Confirm tips appear on the Mac; gather user feedback.

1. **Enforce selectively** — change the rule's chosen channels from **Audit** to **Block** (or

**Block with override / Warn**) for the channels compliance approved, and set the policy to `Enable`: `Set-DlpCompliancePolicy -Identity 'Endpoint DLP - macOS - Pilot' -Mode Enable`. Start with the **highest-risk, lowest-friction** channel (often USB) rather than all at once.

1. Re-run `DLPMac-Report.ps1` — confirm blocks/overrides are recorded as intended.

2. **Rollback if needed:** `Set-DlpCompliancePolicy -Mode TestWithoutNotifications` (back to silent

simulation) reverts enforcement immediately on next sync.

### *Step 10 — Expand to production in waves*

1. Create the production twin `Endpoint DLP - macOS - Prod` (or widen the pilot policy's onboarded

scope), starting again in `TestWithoutNotifications` for the newly onboarded Macs.

1. Onboard production Macs in **waves** (Phase 1) so each wave is validated in simulation before it is

enforced.

1. Re-run `DLPMac-Validation.ps1` and `DLPMac-Report.ps1` after each wave; only advance the mode

once the wave passes.

### *Order of operations*

`Onboard Macs (Intune/Jamf: package + system/network extension + Full Disk Access) → confirm onboarded → Endpoint DLP settings (restricted apps, domains, exclusions) → create DLP policy Devices-scoped in TestWithoutNotifications → add egress rule with Audit actions → discovery/validation confirm test mode → drive test activities + validate in Activity Explorer → TestWithNotifications → enforce (Block) selected channels on Enable → expand in waves (each starting in simulation)`

# 5. Testing Guide — Endpoint DLP for macOS (Microsoft Purview)

Run after the Macs are onboarded and the DLP policy is created **in test/simulation (Audit) mode** (Implementation Phases 1–3), and again after each mode advance and expansion wave. Back each test with `DLPMac-Discovery.ps1` / `DLPMac-Validation.ps1` / `DLPMac-Report.ps1` and the Purview portal. **Use non-production test content** that matches the in-scope SIT — never real regulated data.

## Test 1 — Macs are onboarded and reporting

**Do:** Run `DLPMac-Onboarding-Check.ps1`; Purview portal → **Settings** → **Device onboarding** → **Devices**. **Expected:** The pilot Macs appear **onboarded/active**; the system/network extension is allowed and Full Disk Access is granted on the device.

## Test 2 — DLP policy is Devices-scoped and in test/simulation mode

**Do:** Run `DLPMac-Discovery.ps1`. **Expected:** `Endpoint DLP - macOS - Pilot` exists, targets the **Devices (Endpoint)** location **only**, and **Mode = TestWithoutNotifications** (not `Enable`). Anything already in `Enable` is flagged for review.

## Test 3 — The egress rule is present with the intended channels

**Do:** Run `DLPMac-Validation.ps1`. **Expected:** Rule `macOS Egress - Sensitive Content` exists with the agreed **conditions** (SIT/label) and the intended egress restrictions (cloud/browser, USB, network share, clipboard, print, restricted apps), all at **Audit** for the pilot; the referenced restricted-app group `macOS - Restricted Apps` exists.

## Test 4 — USB / removable media egress is audited

**Do:** On a pilot Mac, copy a test file (matching the SIT) to a USB drive. **Expected:** An **audited** Endpoint DLP event for **copy to removable media** appears in Activity Explorer / `DLPMac-Report.ps1`, attributed to the policy + rule. No block yet (Audit mode).

## Test 5 — Network-share copy is audited

**Do:** Copy the test file to a mounted network share. **Expected:** An **audited** "copy to network share" event is recorded for the Mac.

## Test 6 — Cloud / browser upload is audited

**Do:** Upload the test file from an approved browser (Edge) and, if the Purview extension is deployed, from Chrome/Firefox to a sensitive service domain. **Expected:** An **audited** cloud-egress / unallowed-browser event appears. (If Chrome/Firefox produce nothing, confirm the **Purview browser extension** — Requirement 2.8.)

## Test 7 — Clipboard and print are audited

**Do:** Copy sensitive test content to the clipboard (and paste elsewhere); print the test file. **Expected:** **Audited** clipboard and print events are recorded (subject to current macOS support for those channels).

### Test 8 — Restricted-app access is audited

**Do:** Open the test file with an app listed in `macOS - Restricted Apps`. **Expected:** An **audited** "access by restricted app" event is recorded.

### Test 9 — Notifications appear in TestWithNotifications

**Do:** After passing Tests 4–8, advance the policy to `TestWithNotifications` (`Set-DlpCompliancePolicy -Mode TestWithNotifications`) and repeat one egress action. **Expected:** A **policy tip / notification** appears to the user on the Mac; still **no block**.

### Test 10 — Enforcement blocks the selected channel(s)

**Do:** After compliance/legal sign-off, set the chosen channel(s) to **Block** and the policy to `Enable`; repeat the action for an enforced channel and for a non-enforced one. **Expected:** The **enforced** channel is **blocked** (or blocked-with-override) and recorded; non-enforced channels remain audited. `DLPMac-Report.ps1` shows the block/override events.

### Test 11 — Rollback reverts enforcement

**Do:** Set the policy back to `TestWithoutNotifications` and repeat the previously-blocked action. **Expected:** The action is **no longer blocked** (silent simulation restored) on next sync — confirming the documented rollback works.

### Pass/fail gate

| Gate | Requirement |
|---|---|
| **Onboarding pass** | Tests 1–3 pass (Macs onboarded/reporting; policy Devices-scoped in test mode; rule present) |
| **Simulation pass** | Tests 4–8 pass (each in-scope channel produces an audited event on the Mac) |
| **Enforcement pass** | Tests 9–11 pass (notifications, then Block on selected channels, and rollback all behave as intended) |

> *Do not enforce (Block / `Enable`) until the **Simulation pass** gate is green and compliance/records/legal have signed off the enforcement intent. Re-run Tests 4–8 in simulation for every expansion wave before advancing that wave's mode.*

# 6. Troubleshooting Guide — Endpoint DLP for macOS (Microsoft Purview)

Symptom → likely root cause → fix. Use `DLPMac-Discovery.ps1` (policy scope + Mode + egress restrictions), `DLPMac-Validation.ps1` (pass/fail on scope + mode + channels), `DLPMac-Report.ps1` (Activity Explorer evidence), and `DLPMac-Onboarding-Check.ps1` (Intune-side onboarding/extension health) to confirm before and after each fix. Where a control's availability on macOS is **version- or plan-dependent, verify against current Microsoft guidance / your licensing / tenant** rather than assuming.

## A. Onboarding

### *A Mac never shows any DLP activity (policy "does nothing")*

- **Mac not onboarded.** Endpoint DLP only affects **onboarded** devices. **Fix:** confirm the Mac in Purview → **Settings** → **Device onboarding** → **Devices**; run `DLPMac-Onboarding-Check.ps1` to confirm the onboarding profile is deployed in Intune; onboard it (Implementation Phase 1) if absent.
- **System / network extension not approved.** Without the approved extension the agent can't inspect egress. **Fix:** deploy the **System Extensions** allow + **network/content-filter** profile via MDM; on the Mac, **System Settings** → **Privacy & Security** should show the extension allowed.
- **Full Disk Access not granted.** The agent can't see file activity without it. **Fix:** deploy the **PPPC/TCC (Full Disk Access)** profile via Intune/Jamf; verify it is granted on the Mac. These must be **pushed by MDM**, not clicked per device, for a silent rollout.

### *Onboarding package deployed but the Mac still isn't "active"*

- **Profiles not assigned / not yet applied. Fix:** confirm all four payloads (onboarding, system extension, network extension, Full Disk Access) are **assigned to the pilot Mac group** and have applied; force an MDM sync; re-check `DLPMac-Onboarding-Check.ps1`.
- **Unsupported macOS version.** Support is the **three latest major** releases. **Fix:** update macOS; the onboarding check flags clearly old majors (verify the current supported list).
- **MDE pipeline conflict / duplication.** macOS DLP shares onboarding with Defender for Endpoint. **Fix:** if MDE is already present, don't double-deploy conflicting extensions — **verify the shared configuration against current Microsoft guidance for your tenant.**

## B. Policy scope / mode

### *The policy exists but nothing is audited on Macs*

- **Policy not scoped to Devices.** A policy on Exchange/SharePoint/etc. won't touch endpoints. **Fix:** `DLPMac-Validation.ps1` FAILs if the **Devices (Endpoint)** location is missing — add it (or recreate with `-EndpointDlpLocation All` via `DLPMac-Set-Policy.ps1`).
- **Policy Disabled. Fix:** `DLPMac-Discovery.ps1` reports `Enabled`; enable the policy.
- **No rule / rule has no endpoint restrictions. Fix:** confirm `macOS Egress - Sensitive Content` exists with endpoint restrictions (`DLPMac-Discovery.ps1` lists them); add the channels at **Audit**.

### *The policy is unexpectedly blocking users*

- **Policy is in Enable (enforcing) prematurely. Fix (immediate rollback):**

`Set-DlpCompliancePolicy -Identity 'Endpoint DLP - macOS - Pilot' -Mode TestWithoutNotifications` — reverts to silent simulation on next sync. `DLPMac-Validation.ps1` FAILs when a pilot policy is enforcing unexpectedly.

- **A channel was set to Block instead of Audit. Fix:** set the channel's action back to **Audit**

in the rule; only Block after the simulation gate passes and compliance signs off.

## C. Channel coverage

### *Cloud/browser upload isn't caught*

- **Non-Edge browser without the Purview extension.** Service-domain controls on **Chrome/Firefox**

need the **Microsoft Purview browser extension**; Edge is native. **Fix:** force-install the extension (Requirement 2.8), or test cloud egress from Edge for the initial pilot.

- **Sensitive service domains / browser restrictions not set.** These live in **Endpoint DLP settings**.

**Fix:** configure **Sensitive service domains** and **Browser and domain restrictions** (Step 4).

### *A specific channel (clipboard / print / network share) produces no events*

- **Channel not supported on that macOS version, or not enabled in the rule.** macOS supports a

**subset** of Windows channels and it **evolves**. **Fix:** confirm the channel is in the rule (`DLPMac-Validation.ps1` WARNs if an expected channel is missing) and **verify current macOS support** before concluding it's broken.

### *Restricted-app control does nothing*

- `macOS - Restricted Apps` **group missing or empty.** The rule's app control acts on the group in

**Endpoint DLP settings**. **Fix:** create/populate the group (by bundle ID/name); `DLPMac-Validation.ps1` notes the dependency.

## D. Reporting / evidence

### `DLPMac-Report.ps1` *returns no events*

- **No activity driven / not yet propagated.** Endpoint events can lag. **Fix:** drive a test action

with SIT-matching **test** content, wait, and re-run; widen with `-Days 30`.

- **Role lacks Activity Explorer export. Fix:** ensure the operator has the compliance role that

permits `Export-ActivityExplorerData`.

- **Filter field names drifted.** The script retries without the workload filter; if still empty,

**verify current Activity Explorer field names** against Microsoft guidance and review in the portal.

### *Events show but none are tagged macOS*

- **Platform field absent/normalised differently. Fix:** the report notes this; confirm the device

platform in **Activity Explorer** directly (filter by device) rather than relying on the platform tag.

## E. Determinations (not a technical fix)

• **"Should this be blocked / is this regulated?"** These are **compliance, records, privacy, and

legal **determinations, not endpoint-configuration questions.** Fix:** route to the relevant governance function; this solution configures and evidences the control once they decide.

## Escalation checklist

1. `DLPMac-Onboarding-Check.ps1` — Macs managed, supported macOS, onboarding/extension/PPPC profiles present?
2. Purview portal → **Device onboarding** — do the Macs show **onboarded/active**? (No cmdlet for this.)
3. `DLPMac-Discovery.ps1` — policy Devices-scoped, Mode = a **test** mode, rule egress restrictions listed?
4. `DLPMac-Validation.ps1` — PASS on scope + mode + intended channels + restricted-app group present?
5. `DLPMac-Report.ps1` — driven test activity appears as audited events attributed to the policy/rule?
6. Still stuck → capture the symptom and **verify current Microsoft guidance** for macOS Endpoint DLP

support (version + channel + licensing) in your tenant before any fleet-wide change; keep enforcement in **TestWithoutNotifications** until resolved.

# 7. Best Practices — Endpoint DLP for macOS (Microsoft Purview)

Field-proven recommendations for a safe, defensible macOS Endpoint DLP rollout.

## Design

• **Simulation first, always.** Create every endpoint policy in `TestWithoutNotifications`

(silent simulation / Audit). Blocking macOS egress is high-impact — validate what a policy *would* catch before it interrupts anyone. Advance **TestWithoutNotifications** → **TestWithNotifications** → **Enable** deliberately, one gate at a time.

• **Scope the Mac pilot by onboarding, not by guesswork.** Endpoint DLP applies to **onboarded**

devices; control the pilot by onboarding only the pilot Macs first, and keep the policy on the **Devices** location only so you're testing macOS behavior in isolation.

• **Let compliance decide *what*, you deliver *how*.** Which content is regulated, what must be blocked

vs merely audited, and whether an activity is permissible are **compliance/records/privacy/legal** determinations. Configure and evidence the control; don't invent the policy.

• **Detect on agreed signals.** Base rules on the **SITs/labels/conditions** data owners signed off —

not a reflexive "block everything". Over-broad detection creates noise that buries real events.

## Onboarding

- **Push the extensions and Full Disk Access by MDM — never per device.** The macOS **system extension**, network extension**, and** Full Disk Access (PPPC/TCC) **must arrive as MDM profiles (Intune or Jamf). A single missing profile means the Mac onboards but** reports nothing** — the quietest, most dangerous failure mode.

- **Verify onboarding before you trust the policy.** Confirm each pilot Mac in Purview **device onboarding **and drive a test activity that shows up in Activity Explorer** before** assuming the policy works. Run `DLPMac-Onboarding-Check.ps1` to confirm the profiles are deployed.

- **Reuse the Defender pipeline where present.** macOS Endpoint DLP shares onboarding with Microsoft Defender for Endpoint; if MDE is already deployed the extensions may be shared — **verify against current Microsoft guidance for your tenant** rather than double-deploying.

- **Stay on supported macOS.** Support is limited to the **three latest major macOS versions** — keep the fleet current, and re-verify channel support after a major macOS release.

## Policy and endpoint settings

- **Keep Endpoint DLP settings and rules in lockstep.** The rule's app restriction references the `macOS - Restricted Apps` group and its cloud control references your service-domain groups — if the group/domain settings drift, the rule acts on the wrong thing. Run `DLPMac-Validation.ps1` on every change.

- **Mind the browser boundary.** Edge is covered natively; **Chrome/Firefox need the Purview extension** for service-domain controls. If the extension isn't deployed, either deploy it or scope cloud-egress testing to Edge and say so — don't assume silent coverage.

- **Exclude paths with intent.** File-path exclusions cut noise but can create gaps; document each exclusion and why. Never exclude a path that would let regulated content leave unseen.

## Operations

- **Pilot on a few, then wave.** Prove onboarding + simulation on a handful of Macs, pass the gates, then expand in waves — each new wave starts in **simulation** before its mode advances.

- **Enforce the highest-risk, lowest-friction channel first.** USB copy is often the cleanest first Block; add channels one at a time, watching `DLPMac-Report.ps1` for false positives and override patterns.

- **Know the one-command rollback.** `Set-DlpCompliancePolicy -Mode TestWithoutNotifications` reverts

enforcement to silent simulation immediately on next sync. Rehearse it so an over-blocking incident is a 30-second fix, not a fire drill.

• **Watch overrides and false positives.** A channel with many user overrides (Block-with-override) or false positives is a signal the detection or the channel choice needs tuning — treat it as data, not noise.

### Automation & security

• **Run read-only first, -WhatIf before -Commit.** `DLPMac-Discovery` / `DLPMac-Validation` / `DLPMac-Report` / `DLPMac-Onboarding-Check` are read-only; `DLPMac-Set-Policy.ps1` defaults to **-WhatIf** and, even with **-Commit**, only ever creates the policy in **test/simulation** mode — enforcement stays a manual, deliberate `Set-DlpCompliancePolicy` step.

• **Least privilege, no secrets.** Connect with the least Purview/Graph scopes each task needs (`Connect-IPPSSession` interactively; Graph read scopes for the onboarding check); never place credentials, tokens, or secrets in scripts or source control.

• **Evidence is the deliverable.** Export Activity Explorer results (via `DLPMac-Report.ps1`) as the audit trail that the control exists and works — this is what makes the macOS DLP posture **defensible** to auditors and regulators.

## 8. Deliverables Summary — Endpoint DLP for macOS (Microsoft Purview)

### Documentation

| Document | Purpose |
|---|---|
| Executive Summary | Business case and what's in the box |
| 1. Solution Overview | The macOS Endpoint DLP model, onboarding (Intune/Jamf), the egress channels, simulation-first rollout, what's exact vs deferred |
| 2. Requirements Guide | Every macOS Endpoint DLP prerequisite with Verify / Implement-if-missing (category Purview GRs referenced, not duplicated) |
| Global Requirements (Purview) | Category-wide Purview prerequisites, published once |
| 3. Implementation Guide | Exact, ordered Purview + Intune + Security & Compliance PowerShell steps: onboard macOS, endpoint DLP settings, DLP policy in test/simulation, pilot, validate, enforce; each object named, pass/fail gates + rollback |
| 5. Testing Guide | Validation procedures with pass/fail gates (onboarding → simulation → enforcement) |

| Document | Purpose |
|---|---|
| 6. Troubleshooting Guide | Onboarding, extension/Full-Disk-Access, policy-scope, channel-coverage, and enforcement issues by symptom → root cause |
| 7. Best Practices | Field-proven recommendations for a safe, defensible macOS DLP rollout |

## PowerShell scripts (Security & Compliance PowerShell / Microsoft Graph)

| Script | What it does | Safety |
|---|---|---|
| `DLPMac-Discovery.ps1` | Endpoint DLP readiness: lists DLP policies scoped to the **Devices** location, each policy's **Mode** (flags anything in `Enable`), and each rule's endpoint egress restrictions; advises verifying macOS onboarding in the portal | Read-only (`Connect-IPPSSession`) |
| `DLPMac-Set-Policy.ps1` | Creates the endpoint-scoped DLP policy **in `TestWithoutNotifications` (simulation)** and the `macOS Egress - Sensitive Content` rule with **Audit** endpoint restrictions; idempotent by name; never creates in `Enable` | Default `-WhatIf`; `-Commit` to write (test mode only) |
| `DLPMac-Validation.ps1` | Validates the policy targets **Devices**, is in a **test/simulation** mode, and the rule includes the intended egress channels and references a present restricted-app group | Read-only |
| `DLPMac-Report.ps1` | Exports **Endpoint DLP** activity from **Activity Explorer** (`Export-ActivityExplorerData`) for the Mac fleet — audited/blocked egress events by activity, app, and rule — the evidence trail | Read-only |

| Script | What it does | Safety |
|---|---|---|
| `DLPMac-Onboarding-Check.ps1` | Scope/health helper (Graph): reports managed **macOS** devices + OS versions from Intune and whether the onboarding / extension / Full-Disk-Access configuration profiles are present; defers definitive onboarding status to the Purview portal | Read-only (`Connect-MgGraph`) |

## What the buyer receives

- This implementation guide as a branded **Word** document and **PDF**.
- All five **PowerShell scripts** in a single ZIP.
- A repeatable process: **Assess** → **Design** → **Configure** → **Pilot (simulate)** → **Validate** → **Expand (enforce).**

## Outcome

macOS endpoints brought into the **Microsoft Purview** DLP program: **onboarded**, watched across the key **egress channels** (cloud/browser, USB, network share, clipboard, print, restricted apps), **piloted in simulation**, **validated** in Activity Explorer, and **enforced** on a controlled scope — a closed Mac blind spot that is configured, evidenced, and **defensible**, with regulatory/records/ legal determinations left to the relevant governance functions.

# Appendix A — PowerShell Scripts

Included in the accompanying ZIP. Each script's header documents full usage.

## DLPMac-Discovery.ps1

**Purpose:** Assesses whether the tenant is ready for macOS Endpoint DLP: enumerates DLP compliance policies scoped to the DEVICES (Endpoint) location, reports each policy's Mode and FLAGS any that are already in Enable (enforcing) rather than a test/simulation mode, and lists each rule's endpoint egress restrictions (cloud/USB/network-share/clipboard/print/restricted-apps). Read-only against Security & Compliance PowerShell — makes no changes. macOS device onboarding status is verified in the Purview portal (no supported cmdlet enumerates onboarded devices). HOW TO RUN powershell -ExecutionPolicy Bypass -File .\DLPMac-Discovery.ps1 PowerShell 5.1+ or 7+. Requires the ExchangeOnlineManagement module; connects with Connect-IPPSSession (interactive modern auth). No credentials are stored in this script. EXPECTED RESULTS A list of endpoint-scoped DLP policies with Mode (TestWithoutNotifications / TestWithNotifications / Enable), a WARNING on any policy already enforcing, and each rule's endpoint restrictions. Confirm macOS devices are onboarded in Purview -> Settings -> Device onboarding before relying on any endpoint policy. .NOTES Purview category — macOS Endpoint DLP. Read-only. Endpoint DLP is E5/E5-Compliance and macOS support is version-dependent — verify your licensing/tenant/macOS version. > [CmdletBinding()] param( [string]$PolicyNameFilter = 'Endpoint DLP - macOS' )

**How to run:** powershell -ExecutionPolicy Bypass -File .\DLPMac-Discovery.ps1 PowerShell 5.1+ or 7+. Requires the ExchangeOnlineManagement module; connects with Connect-IPPSSession (interactive modern auth). No credentials are stored in this script. EXPECTED RESULTS A list of endpoint-scoped DLP policies with Mode (TestWithoutNotifications / TestWithNotifications / Enable), a WARNING on any policy already enforcing, and each rule's endpoint restrictions. Confirm macOS devices are onboarded in Purview -> Settings -> Device onboarding before relying on any endpoint policy. .NOTES Purview category — macOS Endpoint DLP. Read-only. Endpoint DLP is E5/E5-Compliance and macOS support is version-dependent — verify your licensing/tenant/macOS version. > [CmdletBinding()] param( [string]$PolicyNameFilter = 'Endpoint DLP - macOS' )

## DLPMac-Onboarding-Check.ps1

**Purpose:** Helps confirm the macOS onboarding prerequisites for Endpoint DLP are in place: reports managed macOS devices and their OS versions from Intune (are they on a supported release?), and lists macOS configuration profiles that look like the onboarding / system-extension / network-extension / Full-Disk-Access (PPPC) payloads this solution requires. Read-only against Microsoft Graph. NOTE: definitive Endpoint DLP onboarding status lives in the Purview portal (Settings -> Device onboarding) — there is no supported cmdlet for it; this is a management-plane health check, not a substitute for the portal confirmation. HOW TO RUN powershell -ExecutionPolicy Bypass -File .\DLPMac-Onboarding-Check.ps1 PowerShell 5.1+/7+. Requires the Microsoft Graph PowerShell SDK; connects with delegated read scopes: DeviceManagementManagedDevices.Read.All, DeviceManagementConfiguration.Read.All. No secrets. EXPECTED RESULTS A count of managed macOS devices with OS-version breakdown (flags anything below the three-latest support window heuristically), and a list of macOS config profiles matching onboarding/extension/ PPPC name patterns. Then confirm the Macs appear onboarded in the Purview portal. .NOTES Purview category — macOS Endpoint DLP. Read-only. macOS supported for the three latest major releases — verify the current supported version list against Microsoft guidance. >

[CmdletBinding()] param( [string]$ProfileNamePattern = 'onboard|defender|dlp|purview|extension|full disk|pppc|content filter' )

**How to run:** powershell -ExecutionPolicy Bypass -File .\DLPMac-Onboarding-Check.ps1 PowerShell 5.1+/7+. Requires the Microsoft Graph PowerShell SDK; connects with delegated read scopes: DeviceManagementManagedDevices.Read.All, DeviceManagementConfiguration.Read.All. No secrets. EXPECTED RESULTS A count of managed macOS devices with OS-version breakdown (flags anything below the three-latest support window heuristically), and a list of macOS config profiles matching onboarding/extension/ PPPC name patterns. Then confirm the Macs appear onboarded in the Purview portal. .NOTES Purview category — macOS Endpoint DLP. Read-only. macOS supported for the three latest major releases — verify the current supported version list against Microsoft guidance. > [CmdletBinding()] param( [string]$ProfileNamePattern = 'onboard|defender|dlp|purview|extension|full disk|pppc|content filter' )

## DLPMac-Report.ps1

**Purpose:** Pulls Endpoint DLP events from Activity Explorer via Export-ActivityExplorerData so you can see exactly what the macOS DLP policy is catching in simulation (audited egress) and, after enforcement, what it is blocking/overriding — by activity, application, and matched rule. This is the audit evidence that the control exists and works. Read-only — makes no changes. HOW TO RUN Last 7 days: powershell -ExecutionPolicy Bypass -File .\DLPMac-Report.ps1 Wider window: ... -Days 30 -OutputCsv .\macos-dlp-activity.csv PowerShell 5.1+/7+. Requires ExchangeOnlineManagement; connects via Connect-IPPSSession (Activity Explorer export requires the appropriate compliance role). No credentials in script. EXPECTED RESULTS A summary of Endpoint DLP activities for the period (grouped by activity type and, where available, platform), and — with -OutputCsv — a CSV of the raw events for the audit file. Empty results usually mean no matching activity yet, macOS not onboarded, or the policy not in scope. .NOTES Purview category — macOS Endpoint DLP. Read-only. Export-ActivityExplorerData paging/filters evolve — verify field names against current Microsoft guidance if a filter returns nothing. > [CmdletBinding()] param( [int]$Days = 7, [string]$OutputCsv, [int]$PageSize = 5000 )

**How to run:** Last 7 days: powershell -ExecutionPolicy Bypass -File .\DLPMac-Report.ps1 Wider window: ... -Days 30 -OutputCsv .\macos-dlp-activity.csv PowerShell 5.1+/7+. Requires ExchangeOnlineManagement; connects via Connect-IPPSSession (Activity Explorer export requires the appropriate compliance role). No credentials in script. EXPECTED RESULTS A summary of Endpoint DLP activities for the period (grouped by activity type and, where available, platform), and — with -OutputCsv — a CSV of the raw events for the audit file. Empty results usually mean no matching activity yet, macOS not onboarded, or the policy not in scope. .NOTES Purview category — macOS Endpoint DLP. Read-only. Export-ActivityExplorerData paging/filters evolve — verify field names against current Microsoft guidance if a filter returns nothing. > [CmdletBinding()] param( [int]$Days = 7, [string]$OutputCsv, [int]$PageSize = 5000 )

## DLPMac-Set-Policy.ps1

**Purpose:** Creates the two objects this solution relies on, SAFELY: (1) a DLP compliance policy scoped to the DEVICES (Endpoint) location in TestWithoutNotifications (silent simulation / Audit) — it is NEVER created in Enable/Block; and (2) a DLP compliance rule ('macOS Egress - Sensitive Content') whose macOS egress channels (cloud upload, USB removable media, network share, clipboard, print, restricted apps) are set to AUDIT. Idempotent: skips objects that already exist by name. Default -WhatIf previews the cmdlets; pass -Commit to create (still test-mode only). Enforcement (Block / Enable) is a deliberate,

separate Set-DlpCompliancePolicy step per the Implementation Guide. HOW TO RUN Preview (default): powershell -ExecutionPolicy Bypass -File .\DLPMac-Set-Policy.ps1 Create (test mode): ... -Commit -SensitiveInformationType 'Credit Card Number' PowerShell 5.1+/7+. Requires ExchangeOnlineManagement; connects via Connect-IPPSSession (interactive modern auth). No credentials are stored in this script. EXPECTED RESULTS The policy 'Endpoint DLP - macOS - Pilot' exists in Mode=TestWithoutNotifications and the rule 'macOS Egress - Sensitive Content' exists with Audit endpoint restrictions. Validate with DLPMac-Validation.ps1; drive test activity and review DLPMac-Report.ps1 before enforcing. .NOTES Purview category — macOS Endpoint DLP. Default -WhatIf; -Commit creates in TEST mode only, never Enable. macOS channel support is version-dependent — verify current support before enforcing. > [CmdletBinding(SupportsShouldProcess=$true)] param( [string]$PolicyName = 'Endpoint DLP - macOS - Pilot', [string]$RuleName = 'macOS Egress - Sensitive Content', [string]$SensitiveInformationType = 'Credit Card Number', # replace with the agreed SIT(s) [string]$RestrictedAppGroup = 'macOS - Restricted Apps', [switch]$Commit )

**How to run:** Preview (default): powershell -ExecutionPolicy Bypass -File .\DLPMac-Set-Policy.ps1 Create (test mode): ... -Commit -SensitiveInformationType 'Credit Card Number' PowerShell 5.1+/7+. Requires ExchangeOnlineManagement; connects via Connect-IPPSSession (interactive modern auth). No credentials are stored in this script. EXPECTED RESULTS The policy 'Endpoint DLP - macOS - Pilot' exists in Mode=TestWithoutNotifications and the rule 'macOS Egress - Sensitive Content' exists with Audit endpoint restrictions. Validate with DLPMac-Validation.ps1; drive test activity and review DLPMac-Report.ps1 before enforcing. .NOTES Purview category — macOS Endpoint DLP. Default -WhatIf; -Commit creates in TEST mode only, never Enable. macOS channel support is version-dependent — verify current support before enforcing. > [CmdletBinding(SupportsShouldProcess=$true)] param( [string]$PolicyName = 'Endpoint DLP - macOS - Pilot', [string]$RuleName = 'macOS Egress - Sensitive Content', [string]$SensitiveInformationType = 'Credit Card Number', # replace with the agreed SIT(s) [string]$RestrictedAppGroup = 'macOS - Restricted Apps', [switch]$Commit )

## DLPMac-Validation.ps1

**Purpose:** Confirms the pilot policy is configured correctly and SAFELY: that 'Endpoint DLP - macOS - Pilot' targets the DEVICES (Endpoint) location, is in a TEST/SIMULATION mode (TestWithoutNotifications or TestWithNotifications) rather than Enable, that its rule includes the intended macOS egress channels, and that any restricted-app control references a named app group. Emits PASS/FAIL/WARN lines and an overall gate. Read-only — makes no changes. HOW TO RUN powershell -ExecutionPolicy Bypass -File .\DLPMac-Validation.ps1 Optionally: ... -PolicyName 'Endpoint DLP - macOS - Pilot' -ExpectEnforcing:$false PowerShell 5.1+/7+. Requires ExchangeOnlineManagement; connects via Connect-IPPSSession. No secrets. EXPECTED RESULTS PASS when the policy is Devices-scoped, in a test mode, and the rule covers the intended egress channels. FAIL if it is enforcing unexpectedly, not Devices-scoped, or missing channels. Use the gate before advancing the policy mode. .NOTES Purview category — macOS Endpoint DLP. Read-only. Verify macOS channel support for your version. > [CmdletBinding()] param( [string]$PolicyName = 'Endpoint DLP - macOS - Pilot', [string]$RuleName = 'macOS Egress - Sensitive Content', [string[]]$ExpectedChannels = @('CloudEgress','RemovableMedia','NetworkShare','CopyPaste','Print','UnallowedApps'), [switch]$ExpectEnforcing # set only when you have intentionally moved to Enable )

**How to run:** powershell -ExecutionPolicy Bypass -File .\DLPMac-Validation.ps1 Optionally: ... -PolicyName 'Endpoint DLP - macOS - Pilot' -ExpectEnforcing:$false PowerShell 5.1+/7+. Requires ExchangeOnlineManagement; connects via Connect-IPPSSession. No secrets. EXPECTED RESULTS PASS when the policy is Devices-scoped, in a test mode, and the rule covers the intended egress

channels. FAIL if it is enforcing unexpectedly, not Devices-scoped, or missing channels. Use the gate before advancing the policy mode. .NOTES Purview category — macOS Endpoint DLP. Read-only. Verify macOS channel support for your version. > [CmdletBinding()] param( [string]$PolicyName = 'Endpoint DLP - macOS - Pilot', [string]$RuleName = 'macOS Egress - Sensitive Content', [string[]]$ExpectedChannels = @('CloudEgress','RemovableMedia','NetworkShare','CopyPaste','Print','UnallowedApps'), [switch]$ExpectEnforcing # set only when you have intentionally moved to Enable )